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Reframing

FROM TOKENS TO IMAGES

- ARC transformations—reflection, symmetry, gravity and copying—are inherently visual and physical.
- Most prior systems serialize grids for LLMs or recurrent reasoning models.
- VARC predicts the output grid as per-pixel classification conditioned on a learned task token.

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Visual priors

CANVAS + 2D STRUCTURE

- Place discrete grids on a 64 x 64 canvas with random translation and scale, then embed colors continuously.
- 2 x 2 patchification supplies locality and translation invariance; 2D RoPE preserves spatial structure.
- The canvas produces exponentially many color-patch combinations, reducing symbolic memorization.

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Two-stage learning

OFFLINE + PER-TASK TTT

- Offline training learns 400 ARC-1 tasks plus 1,000 RE-ARC pairs per task from scratch.
- For each unseen task, independently fine-tune using its 2-4 demonstrations.
- Flip, rotate and color-permute into auxiliary tasks, each with its own embedding.

VARC

ARC Is a Vision Problem!

Vision-Centric Abstract Reasoning from Pixels | 2025

Treat ARC puzzles as few-shot image-to-image translation. Visual priors, task-specific test-time training and multi-view voting let a small ViT generalize to unseen abstract transformations.

ARCvisual reasoningtest-time training

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Inference

510 VISUAL VIEWS

- Run random scale/translation views and average pixels mapped to the same grid location.
- Consolidate 510 views by exact-grid majority voting; retain two most frequent for pass@2.
- After adaptation, reasoning is feed-forward image translation rather than a recurrent chain.

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Evidence

- An 18M ViT reaches ARC-1 54.5% and ARC-2 8.3%; visual priors add 27.7 points.
- Ensembling with a 55M U-Net reaches 60.4% ARC-1, near reported 60.2% average human performance.
- It beats from-scratch recurrent HRM/TRM and is orders smaller than leading LLM systems.

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Meaning & limits

- Attention and task embeddings reveal copying, rays, palettes and semantic clusters without language labels.
- ARC-2 remains 11.1% for the ensemble; 510 views and about 70 seconds TTT per task add cost.
- The result complements language reasoning and invites richer visual priors and pretraining.